

Designing Experiments, Measuring Results

A Hands-on AI Research Workshop

Overview

This workshop teaches MTech students the complete workflow for conducting rigorous machine learning research by running well-designed experiments.

Participants will learn how to take a research idea, break it down into testable hypotheses, design controlled experiments, choose appropriate metrics, and run tests that produce trustworthy results. The workshop combines conceptual foundations with hands-on practice, ensuring students leave with both the mental models and practical skills to conduct their own research.

Duration: 2-3 hours

Format: Lecture, group activities, and hands-on experiment design

Audience: MTech students beginning their research journey

Good-to-have: Familiarity with basic ML concepts and Python

The Problem We're Solving

When students start their research projects, they often face a common set of challenges. They know how to train a model and get a loss curve, but they don't know how to answer questions like:

- "How do I prove that my approach is better than existing methods?"
- "What experiments do I need to run to support my hypothesis?"
- "Which metric should I use: accuracy? F1? pass@1? Something else?"
- "How do I know if my results are meaningful or just random noise?"

- "How many times do I need to run my experiment?"
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What Participants Will Learn

Part 1: The Research Mindset

The workshop begins by reframing how students think about research. A research project is not about "building something cool" but about making a claim and providing evidence to support it. Students will learn to distinguish between a project ("I built X") and research ("I proved X works better than Y because of Z").

We will examine the structure of published papers to understand how experienced researchers frame their contributions. By analyzing 2-3 well-known papers (such as ResNet, BERT, or LoRA), students will see how the hypothesis, experiments, and conclusions connect to form a coherent argument.

The key question students will internalize: *"If I believe X is true, what experiment would convince a skeptic?"*

Part 2: Designing Experiments

This is the core of the workshop. Students will learn a systematic approach to experiment design that they can apply to any research problem.

Understanding Variables

Every experiment involves three types of variables. Independent variables are what you change (model size, learning rate, architecture). Dependent variables are what you measure (accuracy, latency, memory usage). Control variables are what you keep fixed to ensure a fair comparison (dataset, random seed, compute budget). Students will practice identifying these variables in real research scenarios.

Building Experiment Grids

Once variables are identified, students will learn to construct experiment grids structured tables that map out all the runs needed to test a hypothesis. We'll cover how to prioritize which experiments to run first, especially when compute resources are limited.

Baselines and Ablations

A result is meaningless without context. Students will learn why every experiment needs baselines (what happens if we do nothing? what's the current best approach?) and how ablation studies help isolate the contribution of individual components. We'll examine examples of weak baselines in published papers and discuss how to avoid this trap.

Reproducibility

Finally, students will learn the practical requirements for reproducible research: setting random seeds, versioning code and dependencies, using configuration files instead of hardcoded values, and logging everything needed to recreate results.

Part 3: Choosing the Right Metrics

Metrics are how you measure success, and choosing the wrong metric can invalidate your entire research. This section helps students understand which metrics to use and why.

Classification Metrics

For classification tasks, students will learn when to use accuracy, precision, recall, and F1-score. We'll discuss why accuracy can be misleading for imbalanced datasets, and how to choose between precision and recall based on the cost of different errors.

Generation Metrics

For tasks involving text or code generation, the landscape is more complex. Students will learn about **pass@1** and **pass@k**, **BLEU** and **ROUGE**, and functional correctness.

Part 4: Running Tests Properly

Choosing the right metric is only half the battle. Students also need to know how to run tests in a way that produces reliable results.

Why one run isn't enough

ML training involves randomness like weight initialization, data shuffling, dropout. Students will learn why running experiments multiple times (typically 3-5 seeds) is essential, and how to report results as mean $\pm$ standard deviation.

Test Set Discipline

Students will learn the proper role of training, validation, and test sets, and why "peeking" at test results during development invalidates your findings.

Benchmarks: Using and Creating

We'll discuss how to use established benchmarks (MBPP, GSM8K, GLUE, ImageNet, etc.) effectively, including their limitations.

Part 5: Common Pitfalls

Before the hands-on session, we'll cover the most common ways experiments go wrong. Recognizing pitfalls like Data Contaminization, overfitting to the test set, picking the right results, and understanding weak baselines.

Part 6: Hands-on Session

The final session is a hands-on walkthrough where students see the complete pipeline in action. Starting from a simple research question, we will:

1. Formulate a testable hypothesis
2. Identify the variables and design an experiment grid
3. Select appropriate metrics for the task
4. Walk through the code structure for running experiments
5. Run tests across multiple configurations
6. Generate a results table with mean and standard deviation
7. Interpret what the results do and don't tell us

Students will have access to all code and can adapt it for their own projects.
